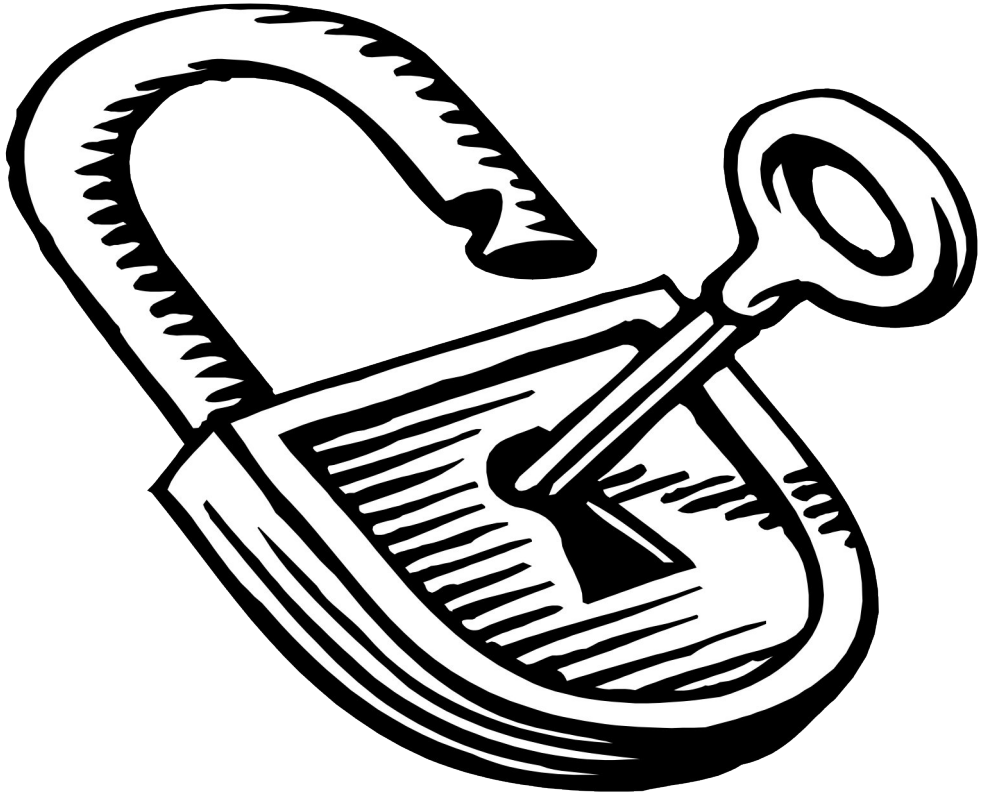
Network & Information Security

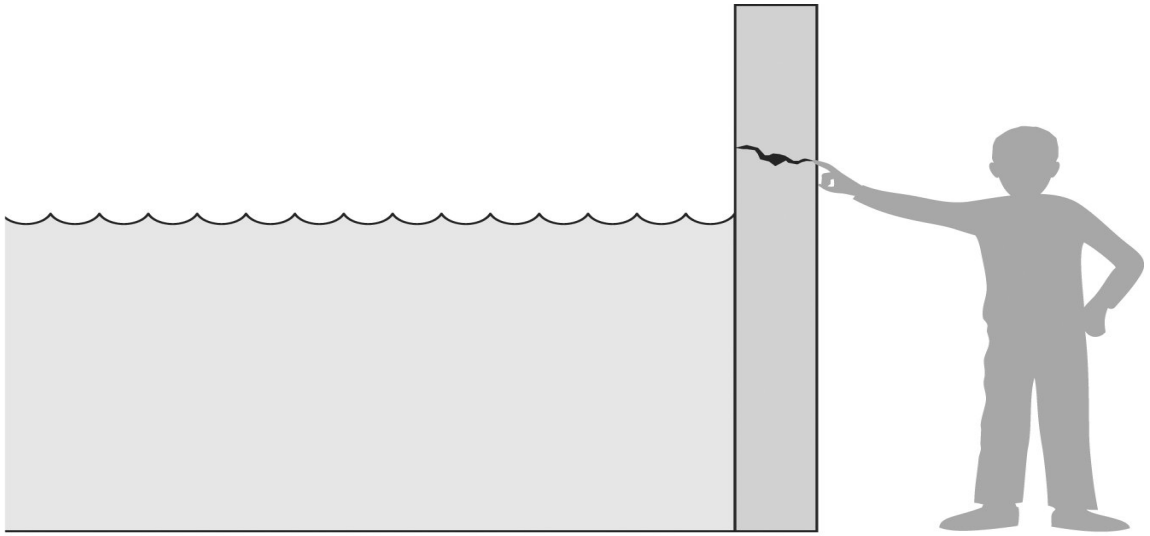


What we will cover?

- Vulnerabilities, threats, security Goals, and methods of defense
- cryptography
- Access Control and security protocols in use for Access Control
- securing computer systems
  - Program security
  - Network security
  - IDS & Firewalls
- Security in networks
- Protocol vulnerabilities

Vulnerability, Threat and Control

- A vulnerability is a weakness in the security system, in procedure, design, or implementation that might be exploited to cause loss or harm
- A threat to a computer system is a set of circumstances that has the potential to cause loss or harm
- Control is an action, device, procedure, or technique that removes or reduces a vulnerability
- A threat is blocked by control of a vulnerability



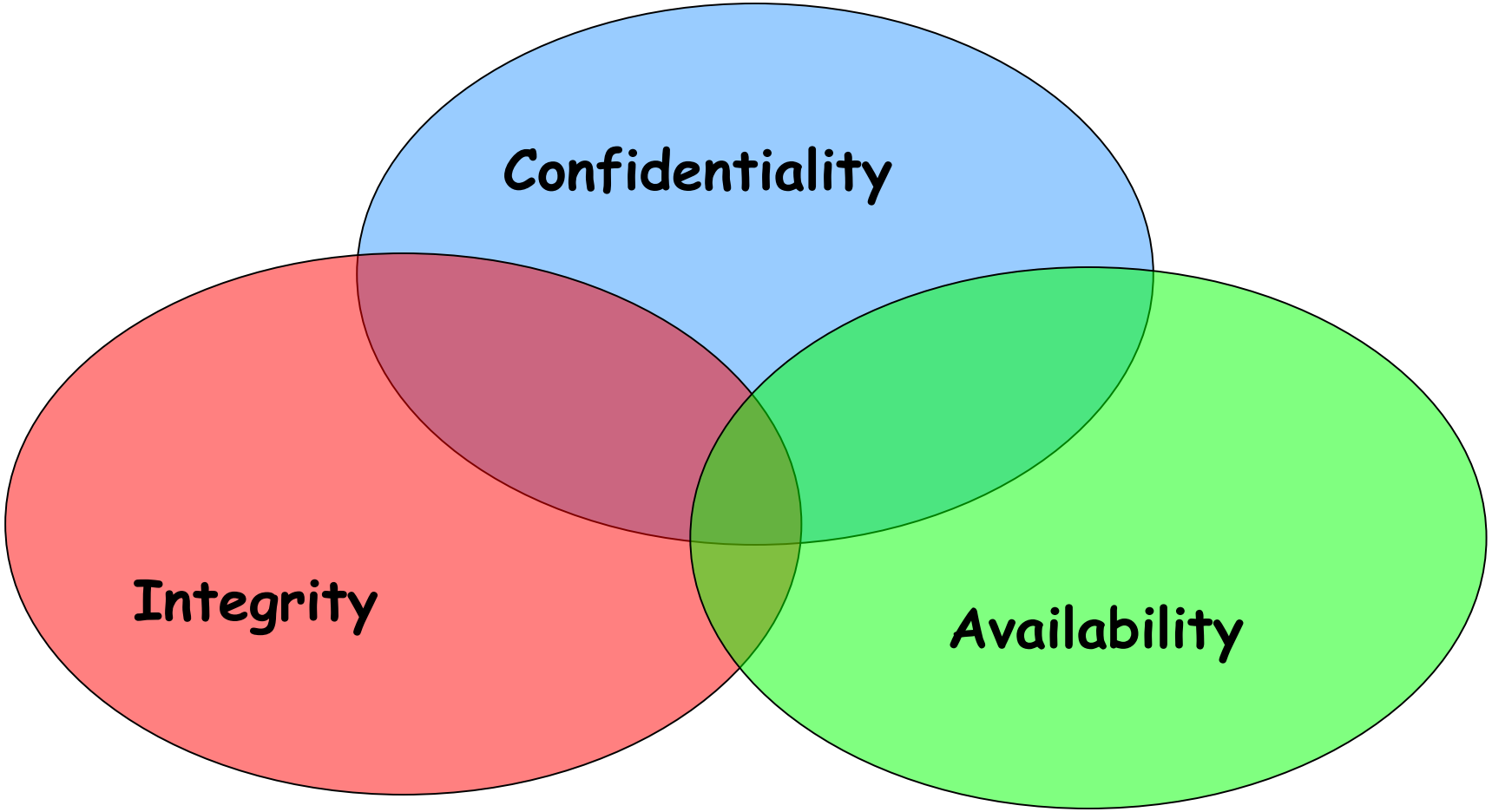
Pfleeger/Pfleeger Fig. 01-01

Figure Threats, Controls, and Vulnerabilities.

Attacks, Services and Mechanisms

- **Security Attack:** Any action that compromises the security of information.
- **Security Mechanism:** A mechanism that is designed to detect, prevent, or recover from a security attack.
- **Security Service:** A service that enhances the security of data processing systems and information transfers. A security service makes use of one or more security mechanisms.

Security Goals



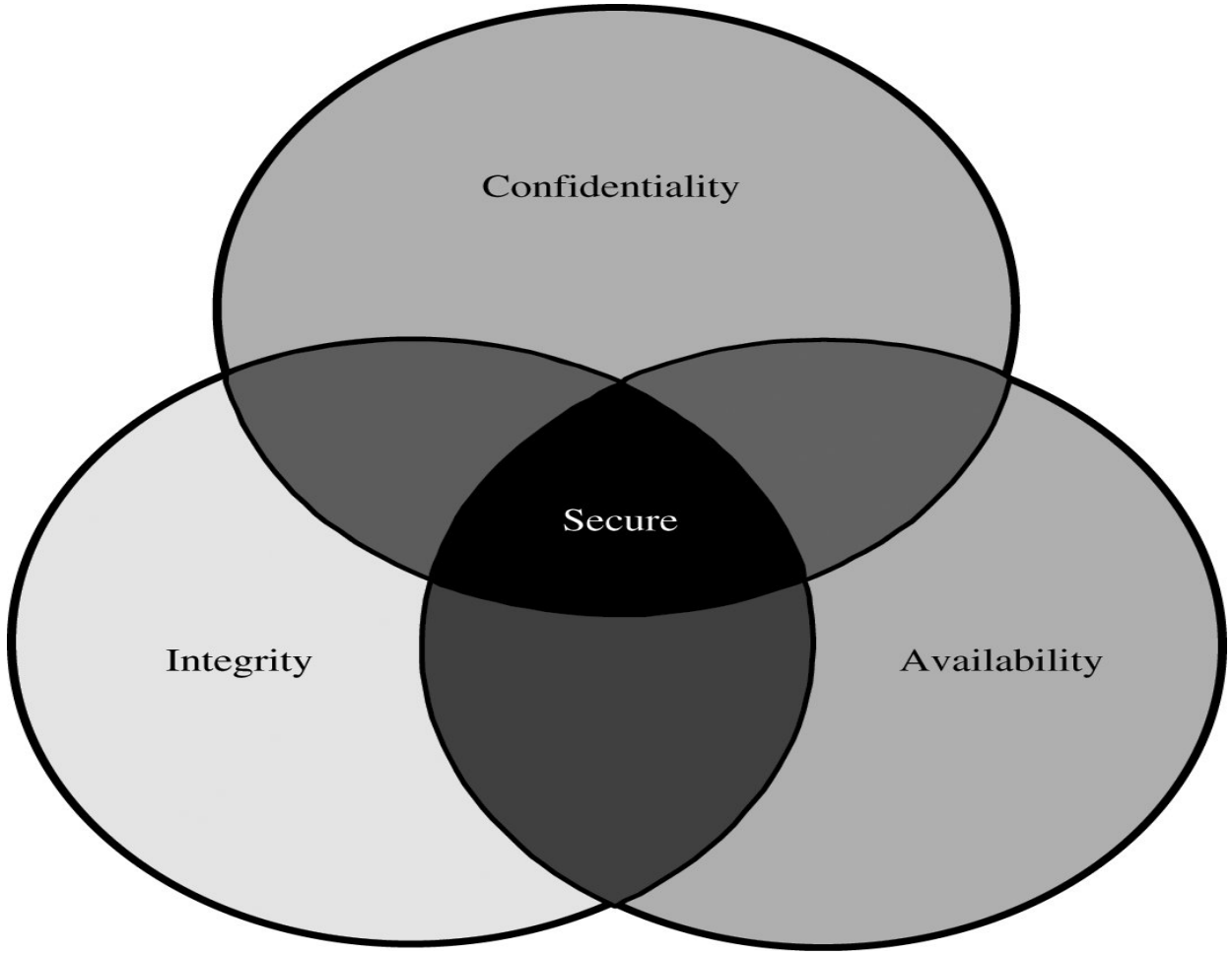


Figure Relationship Between Confidentiality, Integrity, and Availability.

Confidentiality

- It ensures that computer-related assets are accessed only by authorized parties
- Access means reading, viewing, printing, or simply knowing that a particular asset exists
- It is sometimes also called secrecy or privacy

Integrity

- It means that assets can be modified only by authorized parties only in authorized ways.
- The integrity of an item is preserved if it is:
  - Precise, accurate, unmodified, modified only in acceptable ways, modified by authorized people, modified by authorized processes, consistent, meaningful and usable.

Availability

- It applies to both data and data processing
- A data item, service or system is available if
  - There is a timely response to our request
  - Fair to all i.e. some requesters are not favored over others
  - Fault tolerant
  - There is controlled concurrency, deadlock management, and exclusive access as required

Security Attacks

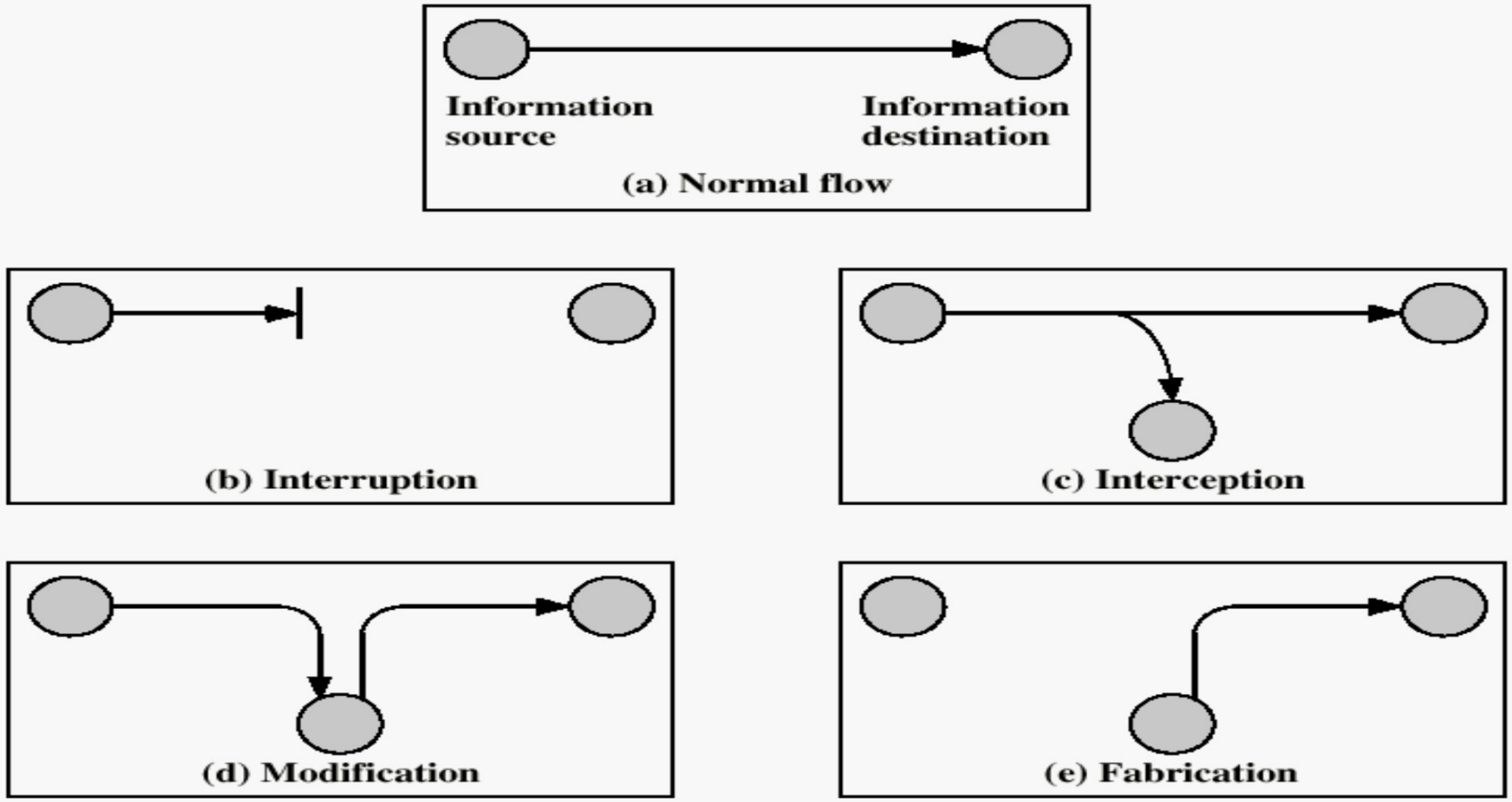
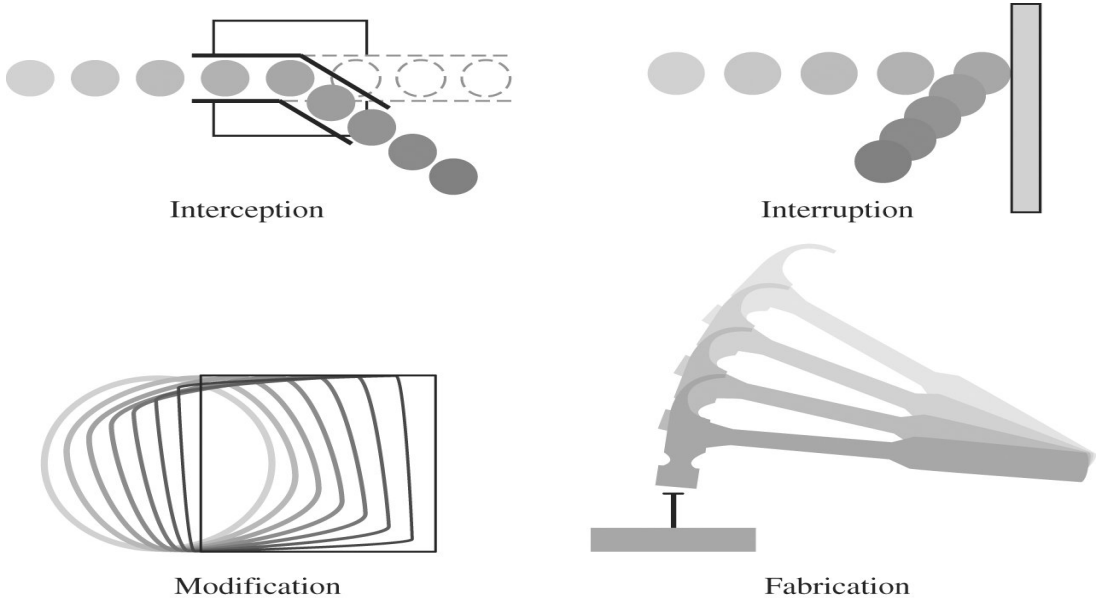


Figure 1.1 Security Threats



Pfleeger/Pfleeger Fig. 01-02

Figure System Security Threats.

Security Attacks

- **Interruption:** This is an attack on availability
- **Interception:** This is an attack on confidentiality
- **Modification:** This is an attack on integrity
- **Fabrication:** This is an attack on authenticity

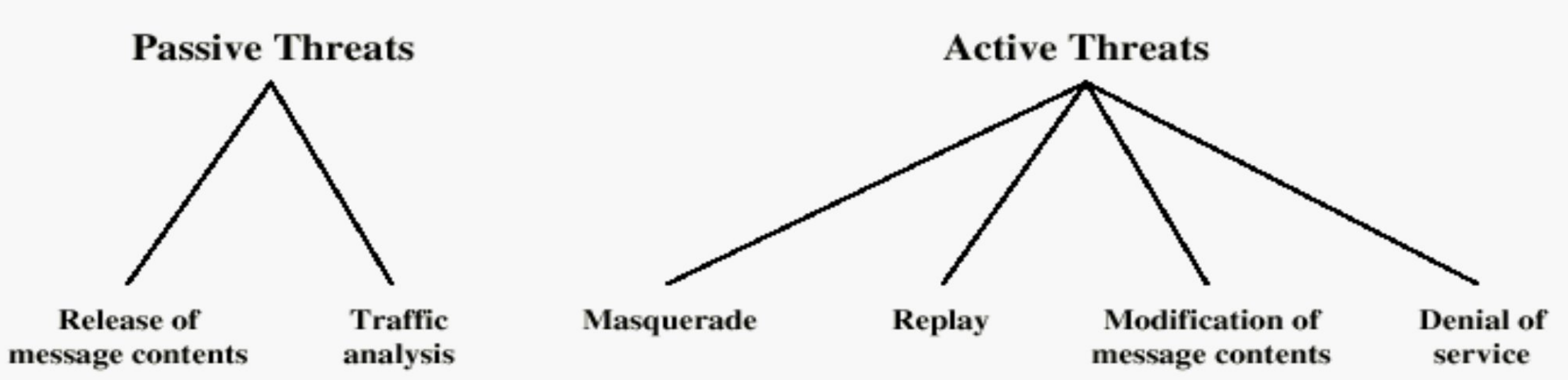


Figure 1.2 Active and Passive Security Threats

Method, Opportunity and Motive

- Method : the skills, knowledge, tools and other things with which to be able to pull off the attack
- Opportunity : the time and access to accomplish the attack
- Motive : a reason to want to perform this attack against this system

DENY ANY OF THESE THREE THINGS AND

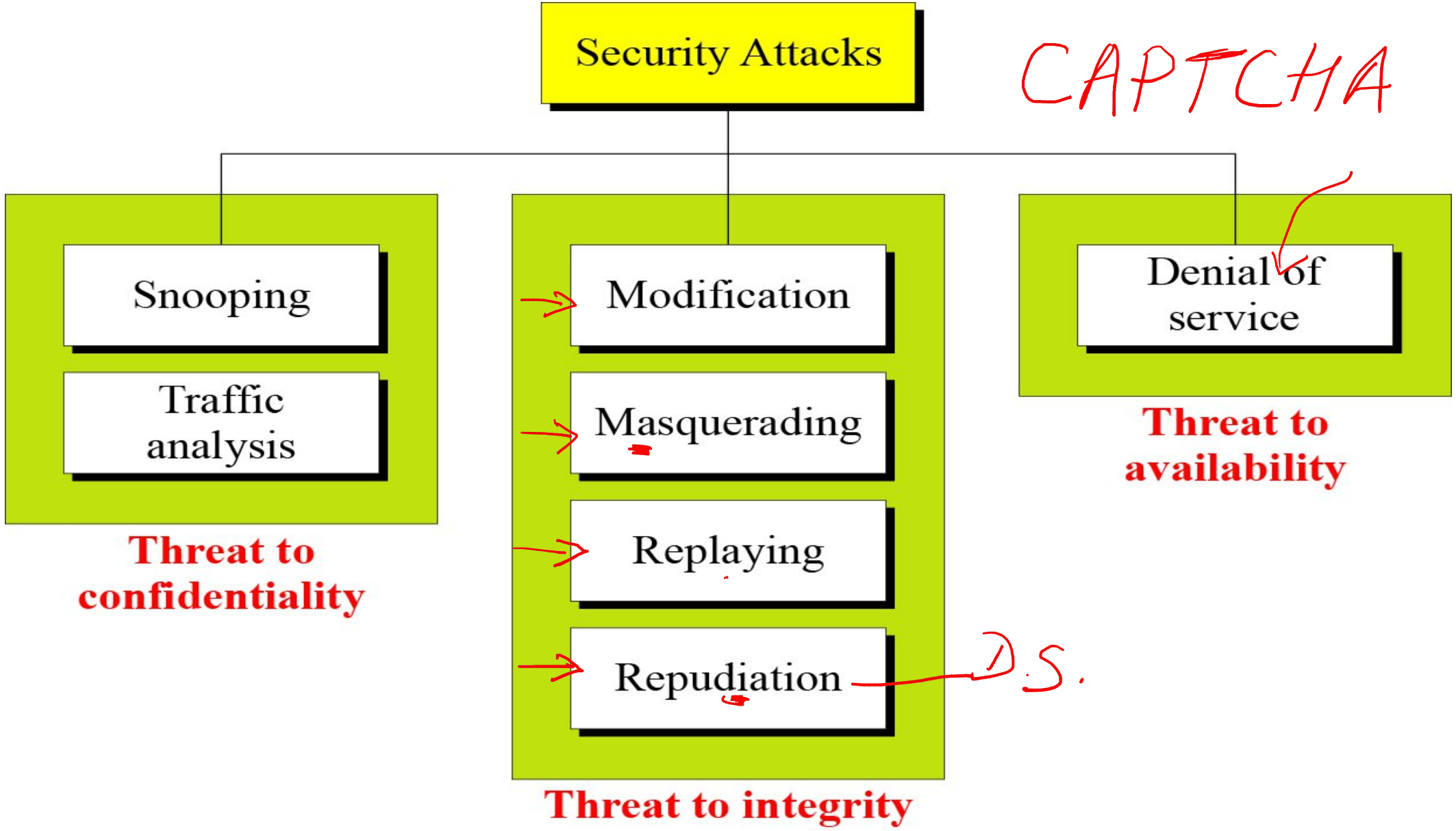
ATTACKS WILL NOT OCCUR

Secure Systems < C, I, A

Attacks

- Cryptanalytic Attacks
  - Exploit mathematical weakness of cryptographic algorithm

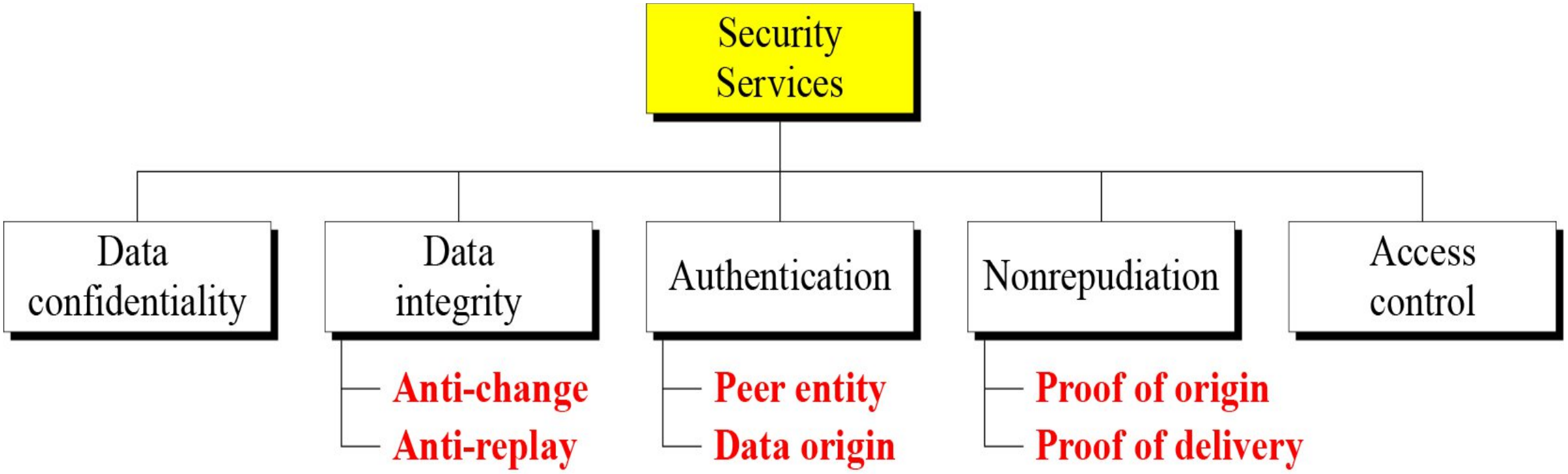
- Non-cryptanalytic Attacks
  - Threats to goal of security


Security Attacks
CAPTCHA
Snooping
Traffic analysis
Threat to confidentiality
Modification
Masquerading
Replaying
Repudiation
D.S.
Threat to integrity
Denial of service
Threat to availability

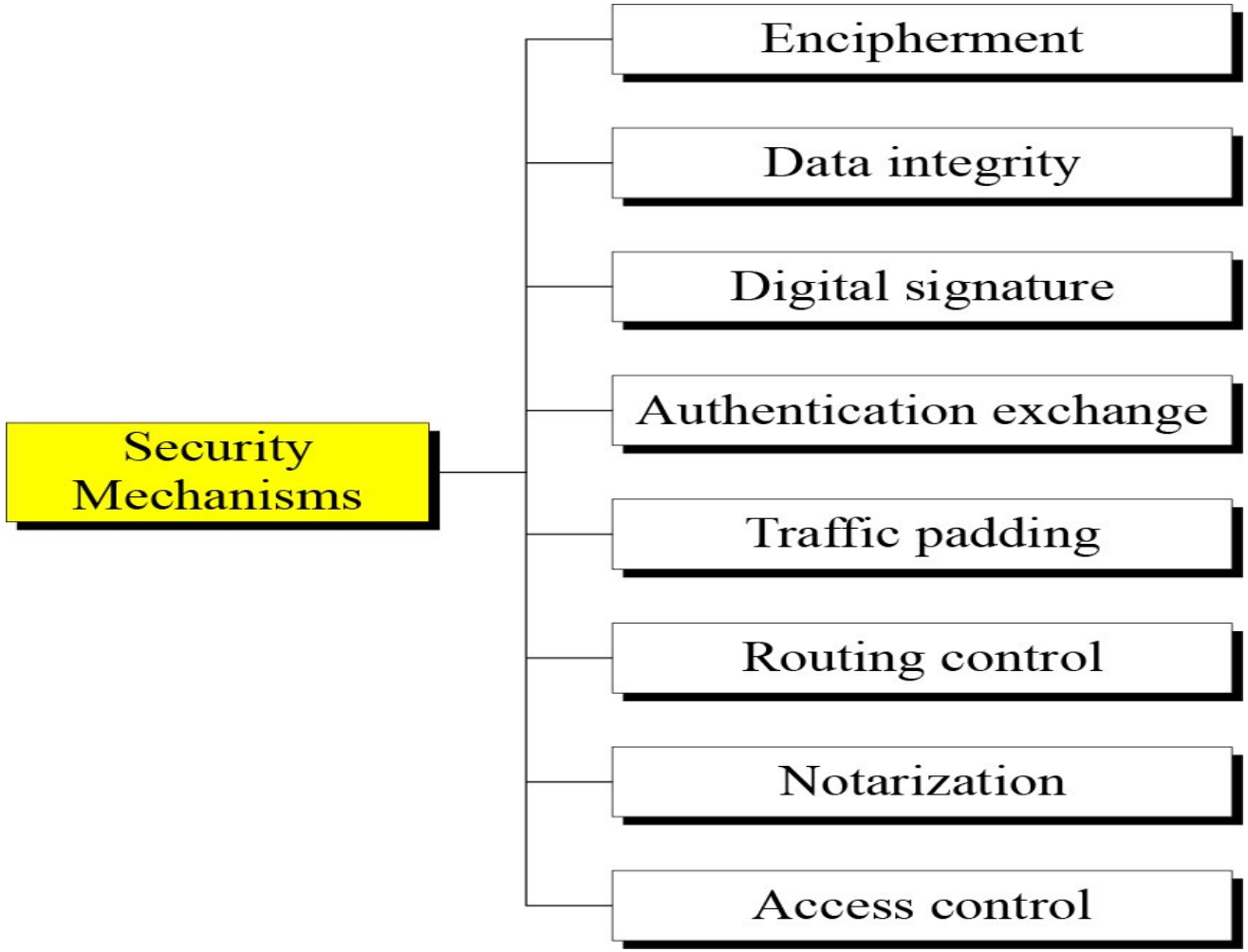
Security Services

- Confidentiality (privacy)
- Authentication (who created or sent the data)
- Integrity (has not been altered)
- Non-repudiation (the order is final)
- Access control (prevent misuse of resources)
- Availability (permanence, non-erasure)
  - Denial of Service Attacks
  - Virus that deletes files

Security Services



Security Mechanism



Relation between Services and Mechanisms

Table

Security Service	*Security Mechanism*
Data confidentiality	Encipherment and routing control
Data integrity	Encipherment, digital signature, data integrity
Authentication	Encipherment, digital signature, authentication exchanges
Nonrepudiation	Digital signature, data integrity, and notarization
Access control	Access control mechanism

Vulnerabilities

- Hardware vulnerabilities
- Software vulnerabilities
  - Software deletion
  - Software modification
    - Viruses etc.
  - Software theft
    - Unauthorized copying etc.
- Data vulnerabilities

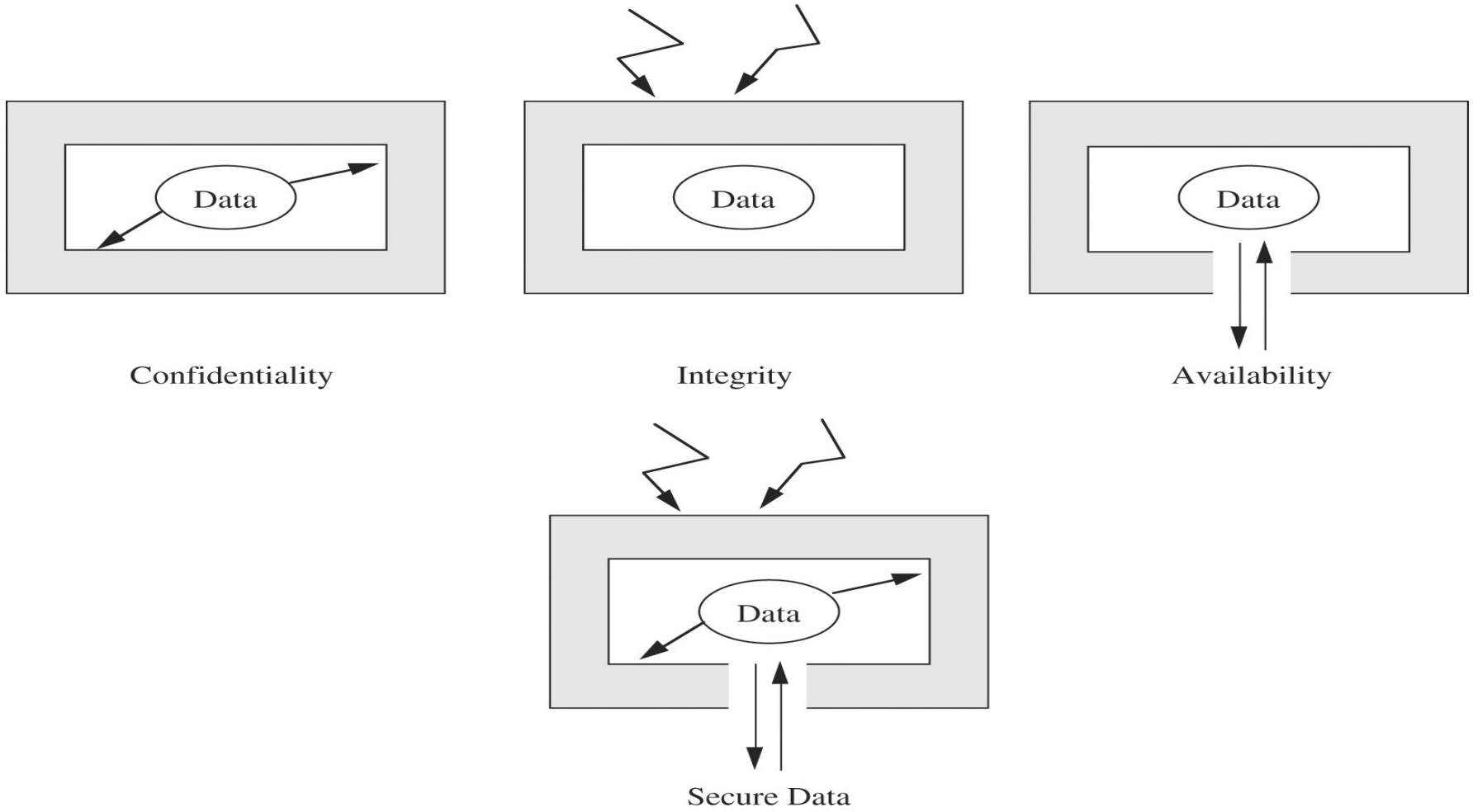


Figure Security of Data.

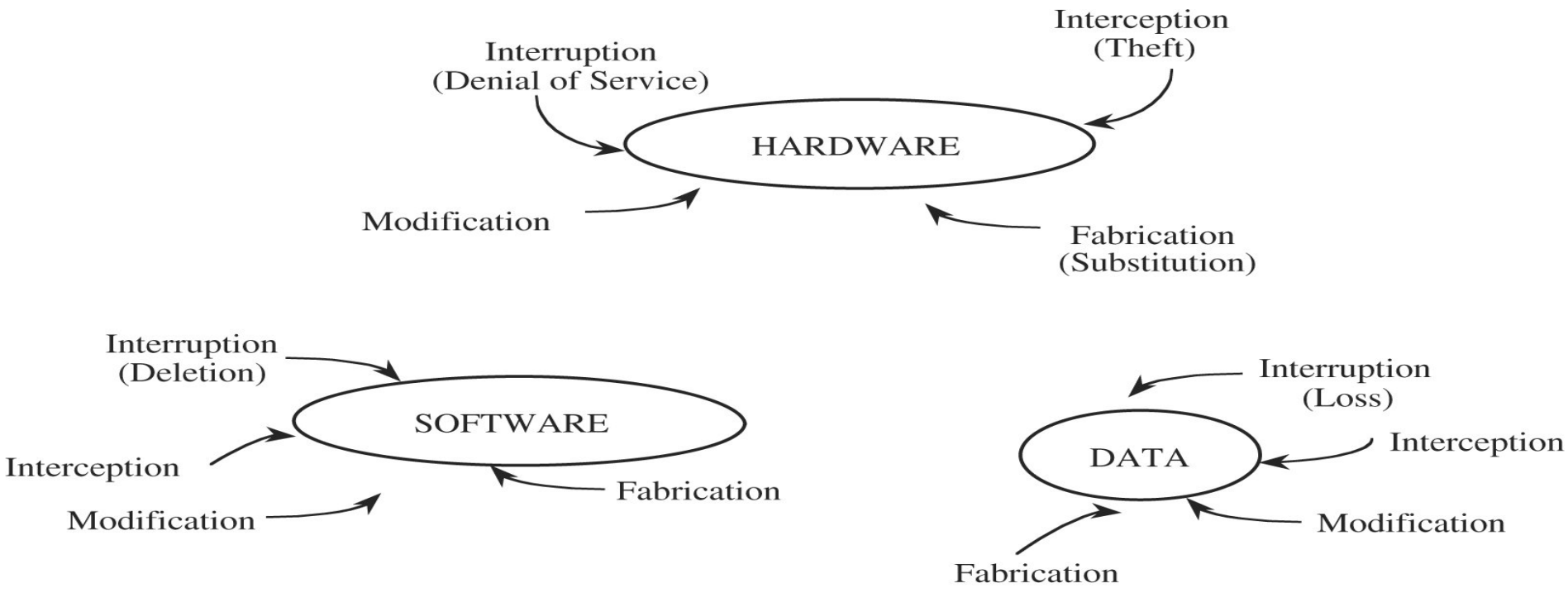


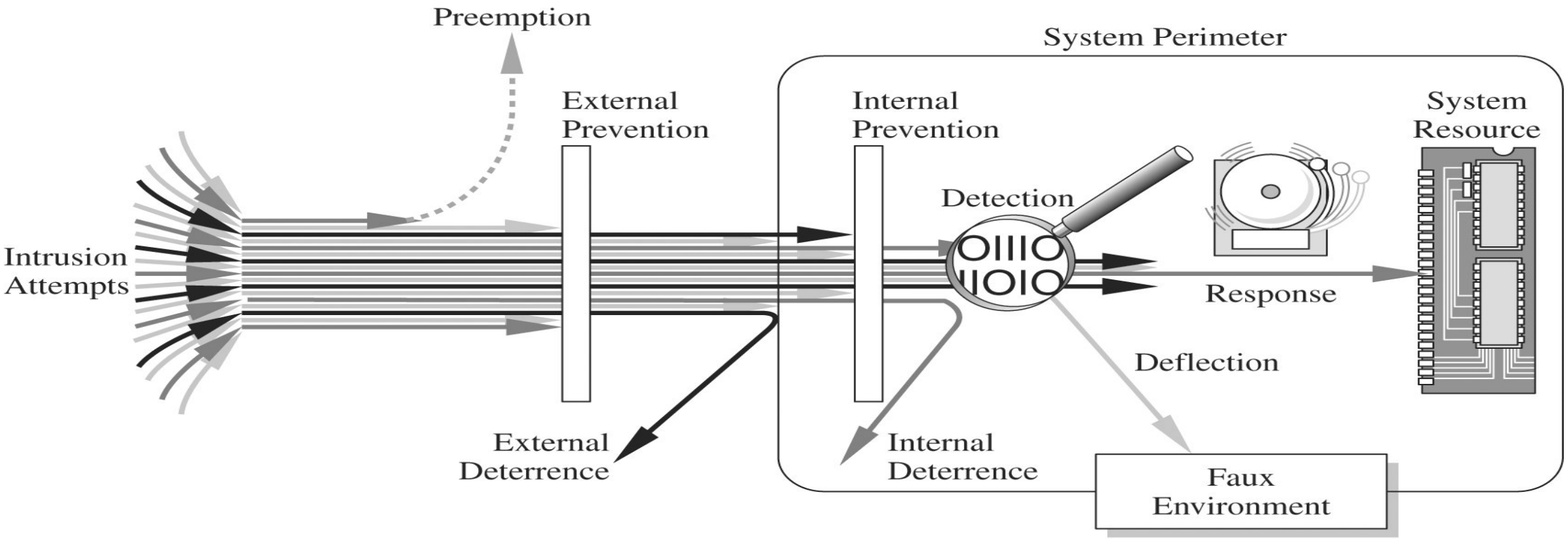
Figure Vulnerabilities of Computing Systems.

Computer Criminals

- Amateurs
  - Personal works
- Crackers
  - Trying to access computing facilities for which they are not authorized
  - The perception that nobody is hurt or even endangered by a little stolen machine time
  - Others attack for curiosity, personal gain, or self-satisfaction
- Career Criminals

Methods of Defense

- *Prevent it*, by blocking the attack or closing the vulnerability
- *Deter it*, by making attack harder if not impossible
- *Deflect it*, by making another target more attractive
- *Mitigate it,* by making its impact less severe
- *Detect it*, either as it happens or some time after the fact
- *Recover* from its effects



Pfleeger/Pfleeger Fig. 01-06

Figure Multiple Controls.

Methods of Defense

- Controls
  - Encryption
  - Hardware Controls
    - Hardware/smart card implementations of encryption
    - Locks or cables limiting access
    - Devices to verify users' identity
    - Firewalls
    - Intrusion detection systems
  - Software Controls
    - Internal program controls,
    - OS and Network system controls
    - Independent control program (anti virus, passwords etc.)
    - Development control
  - Policies and Procedures
  - Physical Controls

Effectiveness of Controls

- Awareness of Problem
  - Highlighting Need of security
- Likelihood of Use
  - They must be efficient, easy to use, and appropriate
- Overlapping Controls
  - Use several different controls, layered defense
- Periodic reviews
  - Judging the effectiveness of control is an ongoing task

Others Exposed Assets

- Networks
  - Network’s lack of physical proximity
  - Use of insecure, shared media
  - Inability to identify remote users positively
- Access
  - Computer time
  - Malicious access
  - Denial of service to legitimate user
- Key People

Your understanding of System Security

Mentimeter

What’s Next

- Encryption overview
- Cryptography in detail